

DAY-1 Hands-On

Level-1 Problem 1: Student Grade Evaluator

Scenario

You are developing a console-based application in .NET 8 for a school. The application should evaluate a student's marks and assign a grade based on predefined rules.

Requirements

- Accept student name and marks (0-100).
- Use if-else statements to determine grade.
- Display grade as A, B, C, D or Fail.
- Handle invalid input using conditional checks.

Technical Constraints

- Use C# (.NET 8 Console Application).
- Use appropriate data types (string, int).
- Use if-else control flow.
- Do not use advanced concepts like classes or LINQ.

Sample Input

Enter Name: Rahul
Enter Marks: 78

Sample Output

Student: Rahul
Grade: B

Expectations

Program should correctly evaluate grade and handle edge cases like marks below 0 or above 100.

Learning Outcome

Understand variables, data types, input/output handling and if-else control statements in C#.

CODE:

```
using System;

class StudentGradeEvaluator
{
    static void Main(string[] args)
    {
        Console.Write("Enter Name: ");
        string name = Console.ReadLine();

        // Check if name is empty or only spaces
        if (string.IsNullOrEmpty(name))
        {
            Console.WriteLine("Invalid name! Please enter a valid name.");
            return;
        }

        Console.Write("Enter Marks: ");
        int marks = Convert.ToInt32(Console.ReadLine());

        if (marks < 0 || marks > 100)
        {
            Console.WriteLine("Invalid marks! Please enter marks between 0 and 100.");
        }
        else if (marks >= 90)
        {
            Console.WriteLine("Student: " + name);
            Console.WriteLine("Grade: A");
        }
        else if (marks >= 75)
        {
            Console.WriteLine("Student: " + name);
            Console.WriteLine("Grade: B");
        }
        else if (marks >= 60)
        {
            Console.WriteLine("Student: " + name);
            Console.WriteLine("Grade: C");
        }
        else if (marks >= 50)
        {
            Console.WriteLine("Student: " + name);
            Console.WriteLine("Grade: D");
        }
        else
        {
            Console.WriteLine("Student: " + name);
            Console.WriteLine("Grade: Fail");
        }

        Console.WriteLine("Press Enter to exit...");
        Console.ReadLine();
    }
}
```

OUTPUT:

```
Enter Name: keerthivasan
Enter Marks: 55
Student: keerthivasan
Grade: D
Press Enter to exit...

C:\Users\keerthi\OneDrive\Desktop\upGrad\DAY-16\HandsOnDay16\HandsOnDay16\bin\Debug\net10.0\HandsOnDay16.exe (process 14292) exited with code 0
(0x0).
Press any key to close this window . . .
```

Level-1 Problem 2: Simple Calculator Using Switch

Scenario

Create a simple calculator application that performs basic arithmetic operations.

Requirements

- Accept two numbers from user.
- Accept operator (+, -, *, /).
- Use switch statement to perform operation.
- Display result.

Technical Constraints

- Use int or double data types.
- Use switch-case statement.
- Handle division by zero.

Sample Input

```
Enter First Number: 10
Enter Second Number: 5
Enter Operator: *
```

Sample Output

Result: 50

Expectations

Correct operator selection and proper validation of inputs.

Learning Outcome

Understand switch statements, arithmetic operators and control flow in C#.

CODE:

```
using System;

class SimpleCalculatorUsingSwitch
{
    static void Main(string[] args)
    {
        Console.Write("Enter First Number: ");
        double num1 = Convert.ToDouble(Console.ReadLine());

        Console.Write("Enter Second Number: ");
        double num2 = Convert.ToDouble(Console.ReadLine());

        Console.Write("Enter Operator (+ - * /): ");
        char op = Console.ReadLine()[0];

        double result = 0;

        switch (op)
        {
            case '+':
                result = num1 + num2;
                Console.WriteLine("Result: " + result);
                break;

            case '-':
                result = num1 - num2;
                Console.WriteLine("Result: " + result);
                break;

            case '*':
                result = num1 * num2;
                Console.WriteLine("Result: " + result);
                break;

            case '/':
                if (num2 == 0)
                    Console.WriteLine("Division by zero not allowed");
                else
                    Console.WriteLine("Result: " + (num1 / num2));
                break;

            default:
                Console.WriteLine("Invalid operator");
                break;
        }

        Console.ReadLine();
    }
}
```

OUTPUT:

```
Enter First Number: 10
Enter Second Number: 10
Enter Operator (+ - * /): /
Result: 1

C:\Users\keerthi\OneDrive\Desktop\upGrad\DAY-16\HandsOnDay16\HandsOnDay16\bin\Debug\net10.0\HandsOnDay16.exe (process 28872) exited with code 0
(0x0).
Press any key to close this window . . .|
```

Level-2 Problem 1: Employee Bonus Calculator

Scenario

Develop a console application that calculates employee bonus based on salary and years of experience.

Requirements

- Accept employee name, salary and years of experience.
- Use if-else and conditional operator.
- Bonus rules:
 - Experience < 2 years: 5% bonus
 - 2-5 years: 10% bonus
 - >5 years: 15% bonus
- Display final salary after bonus.

Technical Constraints

- Use double for salary.
- Use if-else and ternary operator.
- Use proper formatting for currency output.

Sample Input

Enter Name: Aisha
Enter Salary: 50000
Enter Experience: 4

Sample Output

Employee: Aisha
Bonus: 5000
Final Salary: 55000

Expectations

Accurate bonus calculation and correct usage of control statements.

Learning Outcome

Apply conditional logic and arithmetic operations in real-world scenarios.

CODE:

```
using System;
using System.Diagnostics;

class EmployeeBonusCalculator {

    static void Main(string[] args) {

        Console.Write("Enter Name: ");
        String name = Console.ReadLine();

        Console.Write("Enter Salary: ");
        double salary = Convert.ToDouble(Console.ReadLine());

        Console.Write("Enter YOE: ");
        int experience = Convert.ToInt32(Console.ReadLine());

        double bonusPercentage;

        if (experience < 2)
        {
            bonusPercentage = 0.05;
        }
        else if (experience <= 5)
        {
            bonusPercentage = 0.10;
        }
        else {
            bonusPercentage = 0.15;
        }

        double bonus = salary * bonusPercentage;

        //ternary operator example
        double finalSalary = bonus > 0 ? salary + bonus : salary;

        Console.WriteLine("\nEmployee: " + name);
        Console.WriteLine("Bonus: " + bonus.ToString("F2"));
        Console.WriteLine("Final Salary: " + finalSalary.ToString("F2"));
```

```
}  
}
```

OUTPUT:

```
Enter Name: Keerthivasan  
Enter Salary: 20000  
Enter YOE: 3  
  
Employee: Keerthivasan  
Bonus: 2000.00  
Final Salary: 22000.00  
  
C:\Users\keerthi\OneDrive\Desktop\upGrad\DAY-16\HandsOnDay16\HandsOnDay16\bin\Debug\net10.0\HandsOnDay16.exe (process 28448) exited with code 0  
(0x0).  
Press any key to close this window . . .|
```

Level-2 Problem 2: Number Analysis Using Loops

Scenario

Create a .NET 8 console application that analyzes numbers between 1 and N.

Requirements

- Accept a number N from user.
- Use loops to:
 - Count even numbers
 - Count odd numbers
 - Calculate sum of all numbers
- Display results.

Technical Constraints

- Use for or while loop.
- Use int data type.
- Avoid using arrays or collections.

Sample Input

Enter Number: 10

Sample Output

Even Count: 5

Odd Count: 5

Sum: 55

Expectations

Proper loop usage and correct counting logic.

Learning Outcome

Strengthen understanding of loops, counters and accumulators in C#.

CODE:

```
using System;

class NumberAnalysis
{
    static void Main(string[] args)
    {
        Console.Write("Enter Number: ");
        int n = Convert.ToInt32(Console.ReadLine());

        int evenCount = 0;
        int oddCount = 0;
        int sum = 0;

        for (int i = 1; i <= n; i++)
        {
            sum += i;

            if (i % 2 == 0)
                evenCount++;
            else
                oddCount++;
        }

        Console.WriteLine("Even Count: " + evenCount);
        Console.WriteLine("Odd Count: " + oddCount);
        Console.WriteLine("Sum: " + sum);

        Console.ReadLine();
    }
}
```


OUTPUT:

```
Enter Number: 30  
Even Count: 15  
Odd Count: 15  
Sum: 465  
|
```